



Department of Artificial Intelligence and Data Science “Smart Traffic Density & Signal Optimization System”.

Introduction or Abstract

This project uses FastAPI, YOLOv8, OpenCV, and React to create an intelligent traffic management system that detects and counts vehicles from traffic videos in real time. The system dynamically controls traffic signal timing based on vehicle density to reduce congestion and waiting time while providing live monitoring and lane-wise traffic analysis through an interactive dashboard.

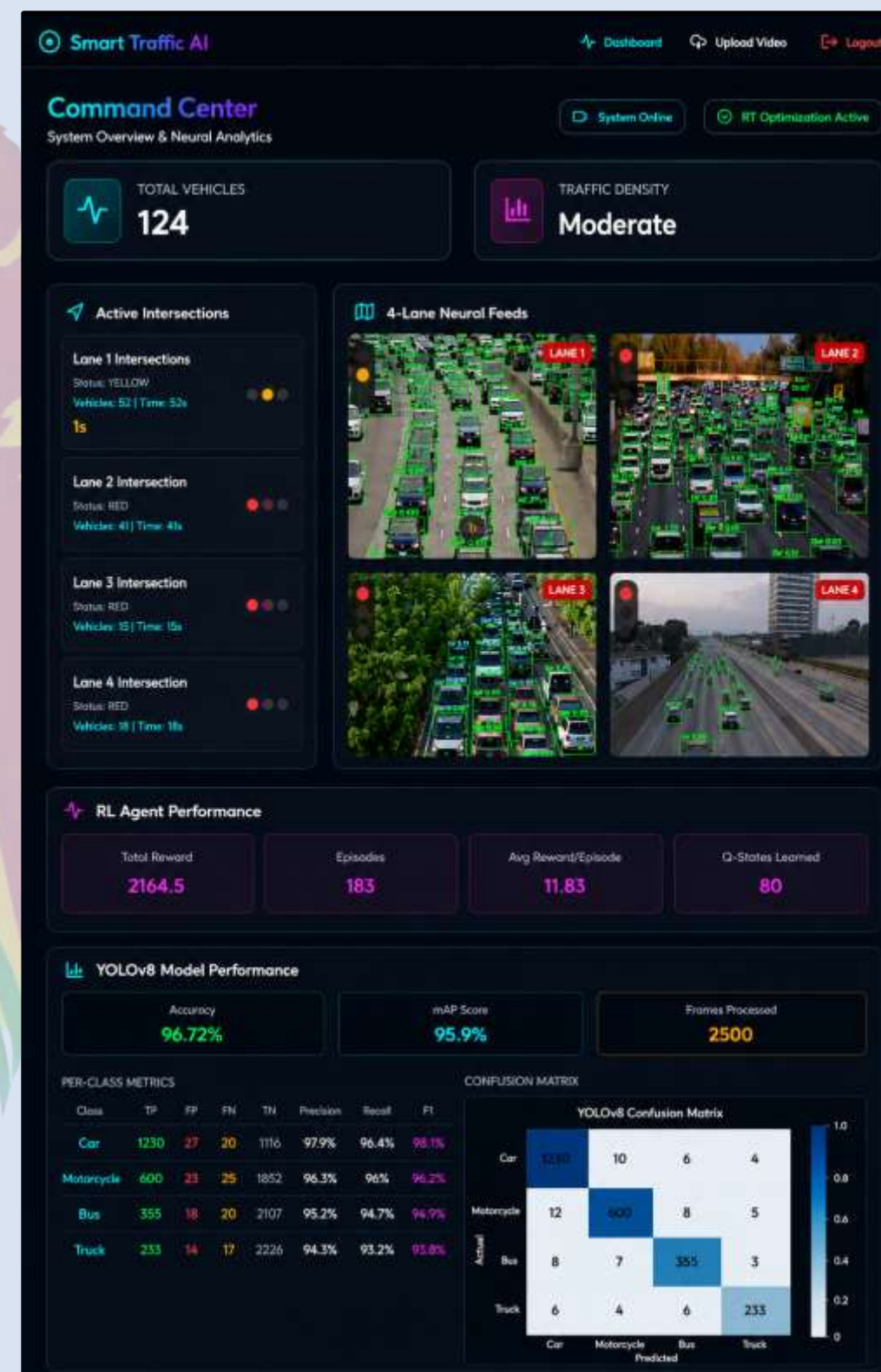
Objective

1. To detect and count vehicles in real time using YOLOv8 and a lightweight FastAPI backend for intelligent traffic monitoring.
2. To dynamically manage traffic signal timing based on vehicle density and provide live traffic analysis, lane-wise statistics, and video monitoring through an interactive dashboard for efficient traffic management.

Methodology



Result



Conclusion

This project successfully integrates a FastAPI backend, YOLOv8 AI-based vehicle detection, OpenCV video processing, and an interactive React dashboard to provide real-time traffic monitoring, vehicle counting, lane-wise analysis, and dynamic traffic signal management, making traffic control more efficient, intelligent, and user-friendly.

Project Team

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